

1500 BCE

The halter yoke

As the use of wheeled vehicles spread, it became necessary to find an efficient way of moving them with animals. The invention of the halter yoke—a set of flat straps stretched across an animal's neck and chest—allowed large weights to be hauled. It also led to the development of light chariots in Egypt, which could be pulled by horses at high speed.



Mummified remains of Pharaoh
Ramses IV (died 1150 BCE)

1000 BCE

Mummification

The Egyptians invented mummification, a way of preserving a dead body by removing the internal organs and wrapping the dried body in linen. Mummifiers reached the height of their skills by 1000 BCE. The process was used mostly for royalty and the wealthy.

1400 BCE

The wood lathe

The lathe, a tool for shaping wood, was invented in Egypt. In its earliest use, one craftsman rotated the piece to be worked using a cord or rope, while a second worker shaped the piece with a sharp tool or chisel.

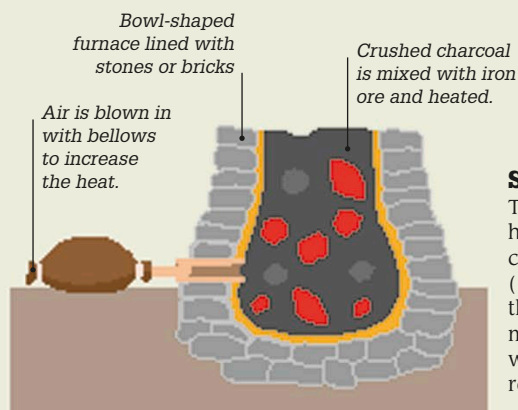
1400

1200

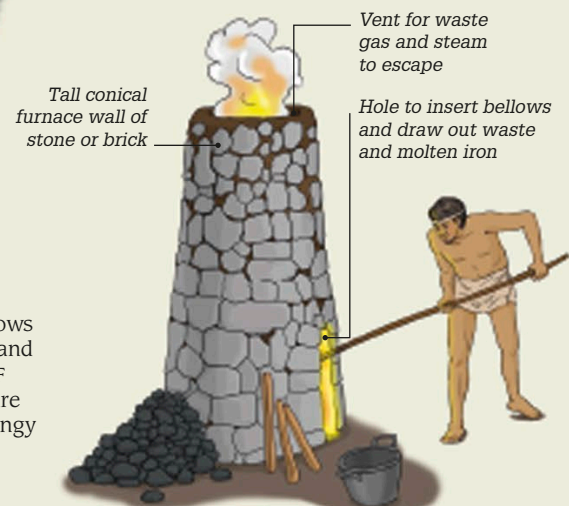
1000

1400–1300 BCE IRON SMELTING

The smelting of iron—extracting iron from iron-bearing ores by heating to a high temperature—was discovered in the Middle East around 1400 BCE, and in India around a century later. The iron produced was much stronger and harder-wearing than bronze, and was used in a variety of tools and weaponry.

**Furnaces develop**

The development of taller shaft furnaces in Roman times enabled more ore and charcoal to be fitted in. The waste slag along with pure molten iron was drawn out at the bottom of the furnace. The mixture of ore and charcoal could be topped up periodically.

**Smelting**

The air pushed in by the bellows heated a mixture of iron ore and charcoal up to around 2010°F (1100°C), at which temperature the iron separated out. A spongy mass of iron was left behind, which became hard when reheated and beaten.